Garrett G. Wen

Ph.D. Candidate
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Education

- **Yale University** — **New Haven, CT, USA**
  Ph.D. candidate, Department of Statistics and Data Science. 09/2022 – Present
  Ph.D. Advisors: Prof. Theodor Misiakiewicz and Prof. Zhou Fan.
  Expected graduation / degree conferral: May 24, 2027.
  M.A. in Statistics (en route), conferred 12/2025.

- **Peking University** — **Beijing, China**
  B.Sc. in Statistics, School of Mathematical Sciences. 09/2018 – 06/2022

Research Experience

- **After42** 2026 – Present
  Core team member.
  Research on AI-generated research questions and expert judgments of research interest.
- **Flatiron Institute, Center for Computational Mathematics (CCM)** **New York, NY**
  Summer Research Associate. 06/01/2026 – 08/14/2026
  Advisors: Dr. Alberto Bietti, Dr. Nikhil Ghosh, and Dr. Denny Wu.
- **The Wharton School, University of Pennsylvania** **Philadelphia, PA**
  Visiting Researcher, Department of Statistics and Data Science. 05/2023 – 07/2023
  07/2024 – 08/2024; 05/2025 – 07/2025
  Remote Research Intern, Department of Statistics and Data Science. 03/2021 – 02/2022
  Research on deep learning theory and peer-review mechanisms.
  Advisor: Prof. Weijie Su.
- **MEGVII**, Beijing, China 02/2022 – 08/2022
  Research Intern, Automated Theorem Proving with Deep Learning.
  Advisor: Dr. Shuchang Zhou.
- **Peking University**, Beijing, China 08/2019 – 02/2022
  Undergraduate Researcher, School of Mathematical Sciences.
  Advisor: Prof. Zhihua Zhang.

Summer Schools

- **Statistical Physics & Machine Learning: Moving Forward**, Cargèse, France 08/2025

- **Princeton Machine Learning Theory Summer School**, Princeton, NJ 06/2023

Research Interests

- Deep learning theory, non-convex optimization, and high-dimensional learning dynamics
- High-dimensional statistics, probability theory, and random matrix theory
- Generative AI, including language-model watermarking and synthetic data
- Mechanism design for scientific evaluation and peer review

Awards & Honors

- Yale Graduate Student Assembly Conference Travel Fellowship 10/2025
- Award for Academic Excellence, Peking University 2019, 2020, 2021, 2022
- **First Prize**, 11th Chinese Mathematics Competition for College Students, National Final (Senior Division) 04/2021
- Guangzhou Pharmaceutical Wanglaoji Scholarship, Peking University 2021
- CreditEase Internet Finance Scholarship, Peking University 2020

Teaching Experience

- **Teaching Fellow (TF) at Yale University** Fall 2024 – Spring 2026
  - S&DS 312/S&DS 612: Linear Models Fall 2024/Fall 2025
    Instructor: Prof. Zongming Ma.
  - S&DS 361/AMTH 361/S&DS 661: Data Analysis Spring 2025
    Instructor: Prof. Brian Macdonald.
  - S&DS 6590: Mathematics of Deep Learning Spring 2026
    Instructor: Prof. Theodor Misiakiewicz. Delivered one course lecture on benign overfitting in February 2026.

Publications

$*$ denotes equal contribution; in (J1), the two co-first authors are listed with Gang Wen first.

Conference and Workshop Papers:

(C2) "Fast Learning Rate Transfer for Gradient Descent in Sketched Linear Regression."
Garrett Wen, Alberto Bietti, Nikhil Ghosh, Theodor Misiakiewicz, Denny Wu.
ICML Workshop on High-Dimensional Learning Dynamics (HiLD), **spotlight**, 2026. Preliminary version of the full paper (P6).

(C1) "PeriodicSketch: Finding periodic items in data streams."
Zhuochen Fan, Yinda Zhang, Tong Yang, Mingyi Yan, **Gang Wen**, Yuhan Wu, Hongze Li, Bin Cui.
In *Proceedings of the 2022 IEEE 38th International Conference on Data Engineering (ICDE)*, IEEE, pp. 96–109, 2022. DOI: 10.1109/ICDE53745.2022.00012.

Journal Papers:

(J3) "Recommending Best Paper Awards via the Isotonic Mechanism."
Garrett G. Wen, Buxin Su, Natalie Collina, Zhun Deng, Weijie Su.
Journal of Machine Learning Research, in press, 2026; page numbers not yet assigned.

(J2) "Self-rankings as a predictor of scientific impact beyond peer review."
Buxin Su, Natalie Collina, **Garrett Wen**, Didong Li, Kyunghyun Cho, Jianqing Fan, Bingxin Zhao, Weijie Su.
Nature Computational Science, 6, Springer Nature, pp. 939–944, 2026. DOI: 10.1038/s43588-026-01039-0.

(J1) "On the Evolutionary of Bloom Filter False Positives—An Information Theoretical Approach to Optimizing Bloom Filter Parameters."
Gang Wen*, Zhuochen Fan*, Zhipeng Huang, Yang Zhou, Qiaobin Fu, Tong Yang, Alex X. Liu, Bin Cui.
IEEE Transactions on Knowledge and Data Engineering, 35(7), IEEE, pp. 7316–7327, 2023. DOI: 10.1109/TKDE.2022.3200045.

Preprints and Technical Reports

** denotes alphabetical order.

(P8) "Can AI Propose Interesting Research Questions?"
Core team: Fan Nie, Zihao Wang, Kaiyue Wen, **Garrett Wen**, Zhaozi Wang, Hai-Yang Wang, Denny Wu, Theodor Misiakiewicz, Joan Bruna, James Zou, Andrea Montanari, Percy Liang.
Contributors: Shuangping Li, Puheng Li, Rui Sun, Renyuan Ma, Sheng Xu, Yiran Zhang, Hugo Latourelle-Vigeant, Ruixiao Wang, Linjun Zhang, Yijun Dong, Mingwei Yang, Alberto Bietti, Anupam Gupta, Tianhao Wang, Theo McKenzie, Shuzhe Zeng, Yanbei Chen, Alexander Philippov, Elias Roland Most, Xiang Li, Juno Kim.
Research blog, After42, 2026. Article; Project.

(P7) "textGrain: Entropy-Calibrated Watermarking for Language Model Text."
Xiang Li, **Garrett Wen**, Xiaohong Chen, Qi Long, Arzav Jain, Florent Joly, Mike Lam, Qingquan Song, Weijie Su.
Technical report accompanying OpenAI's October 5, 2026 announcement, 2026. Technical report; OpenAI announcement.

(P6) "On fast hyperparameter transfer under fixed and diverging time horizons."
Garrett G. Wen, Alberto Bietti, Nikhil Ghosh, Theodor Misiakiewicz, Denny Wu.
Working paper; under review, 2026. OpenReview.

(P5) "Optimal Estimation of Watermark Proportions in Hybrid AI-Human Texts."
Xiang Li, **Garrett Wen**, Weiqing He, Jiayuan Wu, Qi Long, Weijie J. Su.
Minor revision at *Journal of the American Statistical Association*; arXiv:2506.22343, 2025.

(P4) "The sharp SAT/UNSAT phase transition in random ellipsoid fitting."
Theodor Misiakiewicz**, **Garrett G. Wen****.
Submitted to *The Annals of Probability*; arXiv:2608.10184, 2026.

(P3) "Dynamical mean-field limit and replica-symmetric free energy for the orthogonally-invariant SK model."
Zhou Fan**, Theodor Misiakiewicz**, Leda Wang**, **Garrett G. Wen**.
Submitted to *The Annals of Probability*; arXiv:2607.10102, 2026.

(P2) "When does Gaussian equivalence fail and how to fix it: Non-universal behavior of random features with quadratic scaling."
Garrett G. Wen, Hong Hu, Yue M. Lu, Zhou Fan, Theodor Misiakiewicz.
Submitted to *Mathematical Statistics and Learning*; arXiv:2512.03325, 2025.

(P1) "Optimal Detection for Language Watermarks with Pseudorandom Collision."
T. Tony Cai**, Xiang Li**, Qi Long**, Weijie J. Su**, **Garrett G. Wen**.
Preprint arXiv:2510.22007, 2025.

Selected Talks and Presentations

(T3) *International Conference on Statistics and Data Science (ICSDS)*, IMS, Seville 12/2025

(T2) "High-Dimensional Limit Theorems for SGD: Effective Dynamics and Critical Scaling,"
Machine Learning Theory Reading Group, Yale University 09/2024

(T1) "The Isotonic Mechanism for Recommending Best Paper Awards," poster presentation,
Workshop on Incentives in Academia (WINA-24), New Haven 07/2024

Service

- **Reviewer:** Neural Information Processing Systems (NeurIPS), International Conference on Learning Representations (ICLR), ICML Workshop on High-Dimensional Learning Dynamics (HiLD), Journal of Machine Learning Research (JMLR), Annals of Statistics (AOS), and Journal of the Royal Statistical Society: Series B (Statistical Methodology).
- **Committee service:** MA/MS admissions,
Department of Statistics and Data Science, Yale University. 2025

Technical Skills

- **Programming:** Python, R, MATLAB, SQL, C/C++, C#, Java, CUDA.
- **Tools:** PyTorch, TensorFlow, Git, LaTeX, Linux/Ubuntu, Vim, Unity, Origin.

References

- **Prof. Theodor Misiakiewicz**, Department of Statistics and Data Science, Yale University.
theodor.misiakiewicz@yale.edu
- **Prof. Zhou Fan**, Department of Statistics and Data Science, Yale University.
zhou.fan@yale.edu
- **Prof. Weijie Su**, Department of Statistics and Data Science, The Wharton School, University of Pennsylvania.
suw@wharton.upenn.edu
- **Dr. Alberto Bietti**, Center for Computational Mathematics, Flatiron Institute.
abietti@flatironinstitute.org
- **Dr. Denny Wu**, Faculty Fellow, Center for Data Science, New York University; Center for Computational Mathematics, Flatiron Institute.
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